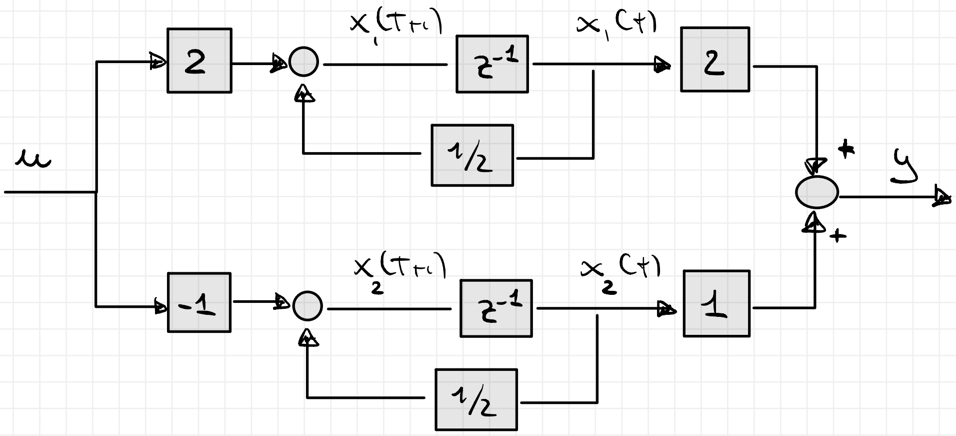
Exercise

Consider the following system:

$$F = \begin{bmatrix} 1/2 & \\ & 1/2 \end{bmatrix} \quad G = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad H = \begin{bmatrix} 2 & 1 \end{bmatrix} \quad D = [0]$$

⇒ Draw the block diagram and discuss the controllability and observability of the system



Analysis: since the input influences both states
since the output contains both states

⇓

We could think that the system is both reachable and observable

⇒ Verify the previous results with the proper tests

▪ Controllability

$$R = \underbrace{\begin{bmatrix} G & FG & \dots & F^{n-1}G \end{bmatrix}}_{n \times n}$$

$$= \begin{bmatrix} 2 & 1 \\ -1 & -1/2 \end{bmatrix} \rightarrow \det R = -1 + 1 = 0 \Rightarrow \text{not full rank!}$$

The system is not controllable!

Observability

$$\mathcal{O} = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{m-1} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1/2 \end{bmatrix}$$

$m \times m$

$$\det \mathcal{O} = 1 - 1 = 0 \Rightarrow \text{NOT FULL RANK!}$$

$$\boxed{\text{THE SYSTEM IS NOT OBSERVABLE!}}$$

IN FACT, USING A SIMPLE VARIABLE TRANSFORMATION:

$$T = \begin{bmatrix} 1 & 0 \\ -1/2 & -1/2 \end{bmatrix} \longrightarrow T^{-1} = -2 \begin{bmatrix} -1/2 & 0 \\ 1/2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & -2 \end{bmatrix}$$

$$x = T \cdot \tilde{x} \Rightarrow \quad T \cdot \dot{\tilde{x}} = F T \tilde{x} + G u$$

$$y = H T \tilde{x} + D u$$

$$\Downarrow$$

$$\begin{cases} \dot{\tilde{x}} = \overbrace{\boxed{T^{-1} F T}}^{\tilde{F}} \tilde{x} + \overbrace{\boxed{T^{-1} G}}^{\tilde{G}} u \\ y = \underbrace{\boxed{HT}}_{\tilde{H}} \tilde{x} + \overbrace{\boxed{D}}^{\tilde{D}} u \end{cases}$$

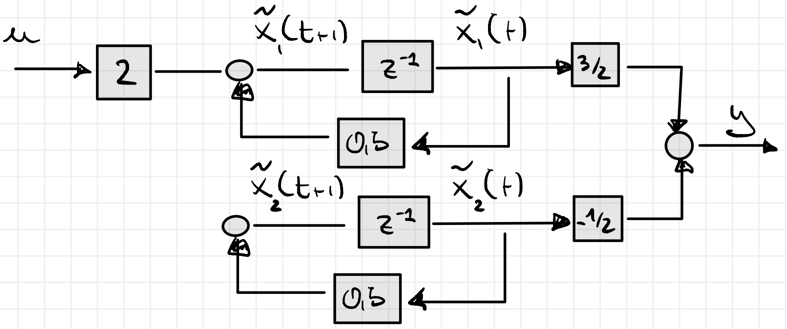
EQUIVALENT SYSTEM DESCRIPTION

$$\tilde{F} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0{,}5 & \\ & 0{,}5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 0{,}5 & \\ & 0{,}5 \end{bmatrix}$$

$$\tilde{G} = \begin{bmatrix} 1 & 0 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\tilde{H} = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 3/2 & -1/2 \end{bmatrix}$$

THE BLOCK DIAGRAM IS:



THIS BLOCK DIAGRAM HIGHLIGHTS THAT ONE SYSTEM STATE IS NOT CONTROLLABLE!

⇒ COMPUTE THE HANKEL MATRIX OF ORDER 2 USING:

1. THE MATRIX R AND O

2. THE SYSTEM IMPULSE RESPONSE

1 $H_x = O_x \cdot R_x \Rightarrow$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1/2 \end{bmatrix} = \boxed{\begin{bmatrix} 3 & 3/2 \\ 3/2 & 3/4 \end{bmatrix}}$$

2 FOR STRICTLY PROPER SYSTEMS:

$$H_x = \begin{bmatrix} w(1) & \dots & w(x) \\ \vdots & & \\ w(x) & \dots & w(2x-1) \end{bmatrix} \rightarrow H_2 = \begin{bmatrix} w(1) & w(2) \\ w(2) & w(3) \end{bmatrix}$$

LET'S COMPUTE THE I.R. SAMPLES, STARTING FROM THE STATE SPACE SYSTEM MODEL:

- $w(0) = 0$
- $w(1) = HG = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \boxed{3}$
- $w(2) = HFG = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & \\ & 1/2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \frac{1}{2} \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \frac{1}{2} \cdot 3 = \boxed{3/2}$
- $w(3) = HF^2G = \frac{1}{4} \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \boxed{3/4}$

$$\Rightarrow \boxed{H = \begin{bmatrix} 3 & 3/2 \\ 3/2 & 3/4 \end{bmatrix}}$$

EQUAL TO THE PREVIOUS ONE